End of Internship Report

Name: Anuranan Bharadwaj

Internship Title: Turbomachinery Software Intern

Company: Concepts NREC, White River Junction, VT 05001

Semester: Spring 2026

Introduction

During the Spring 2026 semester, I worked as a Turbomachinery Software Intern at Concepts NREC. This opportunity originated from direct outreach, where I contacted Jonathan Bicknell, Vice President of Software Products & Engineering, expressing my interest in contributing to the company. Following this, I interviewed with Akshay Bagi, CAE Engineering Software Global Support Manager, under whom I worked during the internship.

Concepts NREC is a leader in turbomachinery design, analysis, and manufacturing, with business segments spanning software development, engineering services, and hardware solutions. Given my academic focus in aerospace engineering, particularly in jet propulsion and gas turbine systems, this role aligned directly with my interests. My work during the internship focused on the intersection of turbomachinery and software development. This experience allowed me to apply theoretical concepts from coursework, such as fluid dynamics, thermodynamics, and turbomachinery principles, to real engineering tools and workflows. At the same time, I gained exposure to industry-level software systems and problem-solving approaches that extend beyond typical academic environments.

Project 1: Automated CFD Testing Framework Development

My primary project involved extending an existing automated testing framework for Axcent to support multiple CFD solvers. The original system was designed to run automated simulations and result comparisons using pbCFD, an in-house solver. My responsibility was to replicate, validate, and expand this framework to integrate an additional solver, FINE/Turbo, created by Cadence Design Systems.

I began by undergoing initial training to understand Axcent's workflow, including CFD simulation setup, execution, and post-processing. Over the first several weeks, I focused on replicating the existing pbCFD-based automation pipeline on my local system. This required working extensively with Python scripts, understanding the internal API (Application Programming Interface) structure, and debugging execution issues to ensure the full pipeline functioned correctly, i.e., from solver execution to post-processing and exporting results into Excel for validation.

Once the pbCFD workflow was successfully operational, I transitioned to integrating FINE/Turbo into the automation framework. This required studying the solver's API and understanding differences in execution flow, data handling, and solver-specific requirements. I collaborated with Robert Ludew, Principal Software Engineer, to navigate API documentation and align implementation with the existing codebase.

To enable compatibility, I modified existing scripts and developed additional functions to handle solver-specific operations, ensuring seamless execution within the automated pipeline. A key challenge encountered during this phase was a persistent gridding error during CFD execution, which was ultimately identified as an API-related issue requiring resolution in a subsequent Axcent software build.

Another issue involved incomplete data generation at the end of simulations. I resolved this by implementing a polling-based monitoring function, which periodically checked solver status and ensured the program only proceeded once the simulation had fully completed. This improved the robustness of the automation pipeline and prevented premature termination of data processing.

Overall, this project strengthened my understanding of CFD workflows, API driven automation, and debugging complex software systems in an engineering environment.

Project 2: Blade Profile Validation and Geometry Consistency Analysis

In parallel, I contributed to a validation study of blade profiles within the Axcent repository under the guidance of Oleg Dubitsky, Director, Corporate Technology/Corporate Fellow. The objective of this project was to evaluate the geometric integrity and consistency of predefined blade profiles when applied to axial turbine configurations.

For this analysis, each blade profile from the Axcent library was applied to an axial turbine model, and key geometric parameters were extracted and recorded. These included camber, maximum thickness and its location, leading and trailing edge radii, and unguided turning angle. The evaluation was performed across multiple configurations, including both clockwise and counterclockwise rotational directions, as well as in SI and British unit systems.

My work involved systematically running these cases, collecting data, and analyzing whether the generated blade geometries remained physically consistent across transformations. Particular attention was given to identifying distortions in blade shape, inconsistencies in leading and trailing edge definitions, and any discrepancies introduced due to unit system conversions.

The study was conducted for both stator and rotor blade configurations, and in total, I analyzed over 20 blade profiles. I compiled the results into a detailed technical presentation, including visualizations such as meridional views, blade cross-sections, and curvature distributions, along with quantitative comparisons of geometric parameters.

This analysis directly contributed to identifying bugs and inconsistencies within Axcent's blade profile implementation. The analysis also supported efforts to ensure that geometry generation remained robust and reliable across different orientations and unit systems.

Conclusion

This internship provided direct exposure to how turbomachinery design and analysis tools are developed, automated, and validated at a commercial level. Through my work on CFD automation and blade profile validation, I applied theoretical concepts from fluid dynamics and turbomachinery to real engineering workflows. I also developed a stronger understanding of how engineering software integrates simulation, data processing, and analysis.

A key takeaway was the importance of systematic debugging and validation when working with complex systems involving multiple solvers and APIs. I learned to isolate issues, verify intermediate outputs, and distinguish between code-level and software-level errors. Overall, this experience strengthened my problem-solving skills, improved my technical expertise in CFD-related workflows, and reinforced the role of software in modern aerospace engineering.